

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

**CM1A - Actuarial Mathematics for
Modelling**

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q. 1) An insurer is comparing two one-year investments for placing a short-term surplus fund.

- Investment A offers interest at 8.4% per annum nominal convertible monthly.
- Investment B offers interest at 8.7% per annum effective.

If the insurer invests Rs. 500,000 today, by how much will the accumulated amount at the end of one year under Investment B exceed that under Investment A?

- A. Rs. 45
- B. Rs. -120
- C. Rs. 165
- D. Rs. -155
- E. Rs. 255

[2 marks]

Q. 2) An investor is considering two bonds, each redeemable at par in 8 years, and each valued at the same yield rate.

- Bond A pays no coupons.
- Bond B pays annual coupons throughout the term.

Which of the following statements is correct?

- A. Bond A and Bond B must have the same Macaulay duration.
- B. Bond A has a lower Macaulay duration than Bond B.
- C. Bond A has a higher Macaulay duration than Bond B.
- D. Macaulay duration cannot be compared unless the coupon rate of Bond B is given.
- E. Bond B has Macaulay duration equal to 8 years.

[2 marks]

Q. 3) A life insurance contract issued to a life aged exactly 40 years will pay INR 250,000 at the end of the year of death, provided that death occurs between exact ages 50 and 65.

Assuming mortality follows the AM92 Ultimate table and the effective annual rate of interest is 4%, calculate the expected present value of the benefit.

- A. INR 10,460
- B. INR 15,820
- C. INR 22,470
- D. INR 30,700
- E. INR 39,610

[2 marks]

Q. 4) An insurer sets up a fund of Rs. 50,000 at time 0 to meet a benefit payment due at time 2 years. The fund earns interest at a force of interest given by

$$\delta(t) = 0.03 + 0.01t, \text{ where } (t) \text{ is measured in years.}$$

Calculate the accumulated value of the fund at time 2.

- A. Rs. 53,607
- B. Rs. 53,945
- C. Rs. 54,081
- D. Rs. 54,164
- E. Rs. 54,665

[2 marks]

Q. 5) Anil is planning to set up a small computer chip factory.

He expects the project to make losses initially due to setup and commissioning costs, followed by positive profits in later years. He asks his actuarial friend to estimate the annual return implied by the following expected profit stream at the end of each year :

(1200),500,450,400

Assuming all cash flows occur at the end of the year, which of the following is closest to the internal rate of return on the project?

- A. 5.55%
- B. 6.10%
- C. 6.38%
- D. 7.12%
- E. 7.50%

[2 marks]

Q. 6) Nimesh wishes to invest in a 3-year government bond with nominal value of Rs. 10,000. The bond pays annual coupons at 7% per annum, payable yearly in arrears, and is redeemed at par at the end of 3 years. The current market price of the bond is Rs. 10,412.60.

The annual effective 1-year spot rate is 4%, and the annual effective 2-year spot rate is 5%. Assuming no arbitrage, calculate the annual effective 3-year spot rate of interest implied by the bond price.

- A. 5.00%
- B. 5.53%
- C. 6.02%
- D. 6.41%
- E. 6.96%

[2 marks]

Q. 7) An asset-liability manager notes that when the annual effective discount rate used for valuing a block of pension payments increases from 5% to 5.5%, the present value falls from INR 840 crores to INR 798 crores.

Assuming a first-order duration approximation, which of the following is the closest estimate of the modified duration of the pension payments?

- A. 8
- B. 9

- C. 10
- D. 11
- E. 12

[2 marks]

Q. 8) A founder is planning to launch a data analytics platform. An amount of Rs. 500 lakhs will be invested at the beginning of each of the first 2 years.

In return, the project is expected to generate net revenue continuously from the beginning of year 1 to the end of year 10 at a rate of Rs. 120 lakhs per annum, increasing continuously at 4% per annum.

Assuming an effective rate of interest of 8% per annum, calculate the net present value of the project at time 0.

- A. Rs. 40.14 lakhs
- B. Rs. 78.03 lakhs
- C. Rs. 112.05 lakhs
- D. Rs. 156.54 lakhs
- E. Rs. 201.34 lakhs

[2 marks]

Q. 9) An insurer reviews two blocks of policies, A and B, at duration 3 years. For the two blocks combined:

- the total gross prospective reserve is Rs. 18,600
- the total expected present value of future gross premiums is Rs. 82,400
- the total expected present value of future benefits and expenses combined is in the ratio 101 : 20 between blocks A and B

If the expected present value of future benefits and expenses combined for block B is Rs. 10,000, calculate the corresponding value for block A.

- A. Rs. 40,500
- B. Rs. 45,500
- C. Rs. 50,500
- D. Rs. 55,500
- E. Rs. 60,500

[2 marks]

Q. 10) A male life aged x and a female life aged y purchase an annuity of Rs. 6,000 per annum, payable annually in arrear so long as both lives survive.

The annuity decreases by 3% each year.

Basis:

- male mortality force: 0.03 for all ages
- female mortality force: 0.02 for all ages
- interest rate: 5% per annum effective

Calculate the present value of the annuity.

- A. Rs. 52,412
- B. Rs. 44,831
- C. Rs. 60,000
- D. Rs. 63,585
- E. Rs. 67,940

[2 marks]

Q. 11) For a life aged 50 exact, you are given:

- annual effective interest rate $i=5\%$
- $A_{50}=0.320$

Assuming benefits are payable at the end of the year of death and annuities are payable yearly in advance, calculate (a due 50)

- A. 13.60
- B. 13.90
- C. 14.10
- D. 14.28
- E. 14.50

[2 marks]

Q. 12) A company provides a one-year employee retention bonus of Rs. 25,000, payable at the end of the year only if an employee aged 40 exact remains in service throughout the year.

An employee may leave service during the year due to:

- death, with dependent probability $q_{40(d)}=0.004$
- resignation, with dependent probability $q_{40(r)}=k$

The expected present value of the bonus is Rs. 22,800, assuming an annual effective interest rate of 5%.

Calculate k.

- A. 0.0322
- B. 0.0384
- C. 0.0426
- D. 0.0468
- E. 0.0501

[2 marks]

Q. 13) An insurer issues 1,000 identical one-year term assurance policies to lives aged 50 exact. Each policy provides a benefit of Rs. 200,000 payable at the end of the year of death.

Basis used for pricing and reserving:

- $q_{50} = 0.004$
- policy reserve at start of year = Rs. 0

During the year, the actual number of deaths is 3.

Ignoring interest, expenses and lapses, calculate the mortality profit for the year.

- A. Rs. 200,000 profit
- B. Rs. 200,000 loss
- C. Rs. 400,000 profit
- D. Rs. 400,000 loss
- E. Rs. 600,000 profit

[2 marks]

Q. 14) Aarav has a life insurance policy that has just completed 3 years. He asks the insurer whether he can stop paying all future premiums and instead make a single payment immediately that would leave the insurer in the same expected financial position as if the policy continued normally.

At the current duration, the insurer estimates:

- expected present value of future benefits = Rs. 48,000
- expected present value of future expenses = Rs. 2,500
- expected present value of future premiums = Rs. 41,200

Which of the following is closest to the single payment that Aarav should make today?

- A. Rs. 6,800
- B. Rs. 7,500
- C. Rs. 8,100
- D. Rs. 9,300
- E. Rs. 50,500

[2 marks]

Q. 15) Rohan pays tuition fees of Rs. 1,50,000 at times 0, 6, 12 and 18 months to complete a 2-year MSc in Actuarial Science with Data Analytics.

After graduating at 24 months, he starts a job paying Rs. 10,00,000 per annum, payable monthly in arrear, with the first salary payment at 25 months. Salary increases by 20% per annum from the start of each subsequent year of employment.

Income tax is 20% of salary, and all salary after tax is available to recover the tuition fee investment.

Assuming an effective interest rate of 10% per annum, calculate the discounted payback period to the nearest month.

- A. 31 months
- B. 32 months
- C. 33 months
- D. 34 months
- E. 35 months

[2 marks]

- Q. 16)** Rajesh, aged 62 exact, is a member of a pension scheme. His wife, Meera, is aged 59 exact. Under the scheme rules, if Rajesh dies within the next 3 years, a lump sum spouse's benefit of Rs. 60,000 will be paid at the end of the year of Rajesh's death, provided that Meera is alive at that time.

Assume that:

- Rajesh's mortality follows the PMA92 Base table
- Meera's mortality follows the PFA92 Base table
- the future lifetimes of Rajesh and Meera are independent
- the effective rate of interest is 4% per annum

Calculate the expected present value of this spouse's benefit.

[5 marks]

- Q. 17)** A company provides a one-year employee benefit for workers aged 45 exact who are currently in active service. During the year, an employee may leave service due to one of the following competing causes:

- death
- ill-health retirement
- resignation

The dependent probabilities of decrement over the year are:

$(d)q_{45}=0.003$, $(i)q_{45}=0.012$, $(r)q_{45}=0.085$

The benefit structure is as follows:

- Rs. 500,000 is paid immediately on death
- Rs. 300,000 is paid at the end of the year on ill-health retirement
- Rs. 50,000 is paid at the end of the year if the employee remains in active service throughout the year
- No benefit is paid on resignation

Assume:

- the force of interest is constant and equivalent to an annual effective rate of 5%
- deaths are uniformly distributed over the year
- ill-health retirements and resignations are treated as occurring effectively at the end of the year
- no other decrements operate

- i) Calculate the expected present value of the benefits payable over the year to an employee aged 45 exact currently in active service. **[4]**

- ii) State briefly why the resignation probability affects the expected present value, even though no benefit is paid on resignation. **[1]**

[5 marks]

- Q. 18)** An insurer has issued 500 one-year term assurance policies to lives aged 50 exact. Each policy provides a sum assured of Rs. 300,000, payable at the end of the year of death.

At the start of the year:

- the reserve per policy is Rs. 0
- a premium of Rs. 1,500 per policy for this year's cover has already been received
- this premium includes an expected expense loading of Rs. 120 per policy, payable at the start of the year
- the expected mortality basis is $q_{50}=0.004q_{\overline{50}}=0.004q_{50}=0.004$
- the pricing basis also allows for a lapse probability of 2%, with lapses assumed to occur uniformly over the year
- no reserve is held at the end of the year for policies that have lapsed

During the year:

- the actual number of deaths observed is 1
- the actual initial expense incurred was Rs. 130 per policy
- no lapses occurred

At the end of the year, the insurer sets up a reserve of Rs. 1,200 for each policy that remains in force.

Consider interest is 5% per annum. Further it is given that actual and expected interest rate is same.

Calculate:

- the expected profit for the year
- the actual profit for the year
- the profit due to mortality
- the profit due to expenses
- the profit due to lapses

[5 marks]

Q. 19) A perpetuity costs 35.48 and makes payments only at the end of even years.

The perpetuity pays:

- 1 at the end of year 2,
- 2 at the end of year 4,
- ...
- n at the end of year $2n$.

After the payment at the end of year $2n$, the payments remain constant at n every 2 years forever.

The annual effective rate of interest is 8%.

- Determine the value of n . **[4]**
- Calculate the value of the perpetuity immediately after the payment at time $2n$. **[1]**

[5 marks]

Q. 20) A finance company grants a loan of Rs. 500,000 to a small business owner. The loan is to be repaid by 6 annual installments in arrear.

The contractual rate of interest is 9% per annum effective.

The borrower agrees that:

- the first 4 instalments will be equal,
- the 5th instalment will be 20% higher than each of the first 4 instalments, and
- the 6th instalment will clear the outstanding balance exactly.

Immediately after the 4th instalment, the outstanding balance is to be Rs. 180,000.

Calculate:

- i) the amount of each of the first 4 instalments. [2]
- ii) the amount of the 5th instalment and the 6th instalment [2]
- iii) the total amount of capital repaid in the first 3 instalments [2]
- iv) the effective annual rate of interest actually earned by the lender if the borrower, instead of making the 6th installment, redeems the loan immediately after the 5th installment - show the calculation. [2]
- v) A borrower repays a loan by equal annual installments. Without calculation, explain how an increase in the rate of interest would affect the outstanding balance immediately after the first instalment, if the instalment amount remained unchanged. [2]

[10 Marks]

Q. 21) An insurer is assessing a proposed underwriting automation platform. The project would be launched immediately and is expected to generate cost savings and additional contribution over the next 5 years.

The expected cash flows are as follows:

- Rs. 8,00,000 at time 0 for software development
- Rs. 4,00,000 at the end of 9 months for integration and testing
- Rs. 1,20,000 at time 0 as additional working capital, recoverable at the end of year 5
- net operating inflows of:
 - Rs. 2,80,000 at the end of year 1
 - Rs. 4,50,000 at the end of year 2
 - Rs. 5,20,000 at the end of year 3
 - Rs. 4,00,000 at the end of year 4
 - Rs. 3,20,000 at the end of year 5
- maintenance expenditure of Rs. 50,000 payable at the start of each of years 2 to 5
- a residual value of Rs. 80,000 at the end of year 5

The insurer uses a hurdle rate of 8% per annum effectively.

Calculate:

- i) the net present value of the project at time 0. [5]
- ii) the discounted payback period. [3]

- iii) whether the project should be accepted on financial grounds, stating any limitations of using only these measures. [2]

[10 marks]

Q. 22) An insurer issues a 3-year endowment assurance to a life aged 50 exact for a sum assured of Rs. 200,000.

The benefit is payable:

- at the end of the year of death if death occurs within the 3-year term, or
- at the end of 3 years if the policyholder survives to that time.

Level annual gross premiums are payable in advance for a maximum of 3 years.

Life follows AM92 Ult.

Expenses:

- initial expense: Rs. 1,000 at issue
- renewal expense: Rs. 100 at the start of each policy year after the first, provided the policy is still in force
- Interest rate: 5% per annum effective

Calculate:

- i) the gross annual premium, using the equivalence principle. [6]

- ii) the gross prospective reserve at the end of policy year 1, immediately after the premium due at that time and the renewal expense then payable have been paid. [4]

[10 marks]

Q. 23) An insurer has launched a 3-year unit-linked insurance policy for policyholders aged 35 exact.

Under the contract:

- an annual premium of Rs. 40,000 is payable in advance at the start of each of the first 3 policy years, provided the policyholder is alive,
- assume all premiums due are paid,
- ignore lapses, surrender benefits and tax.

The death benefit at the end of the year of death is the higher of:

- the unit fund value, and
- Rs. 120,000

You are given:

$q_{35}=0.0015$, $q_{36}=0.0017$, $q_{37}=0.0019$

Unit side

- 70% of the first premium and 95% of each renewal premium are allocated to the unit fund.
- The unit fund earns 6% per annum effective in policy year 1.
- At the end of each policy year, a fund management charge of 1% of the fund immediately before deduction is deducted.

- From policy year 2 onwards, the insurer now expects future unit growth to be 3% per annum effective.

Non-unit side

- The balance of each premium is a non-unit inflow.
- Expenses are Rs. 15,000 at issue and Rs. 3,000 at the start of each renewal year.
- At the end of each policy year, the insurer meets the cost of any excess of death benefit over the unit fund from the non-unit account.
- Interest rate of 7% per annum effective.

i) Calculate the unit fund value at the end of each policy year, after deduction of fund management charges. [4]

ii) Calculate the expected cost of the death benefit guarantee in each policy year:

- per policy in force at the start of that year, and
- per policy issued at outset.

State clearly the difference between these two measures. [4]

iii) Calculate the non-unit profit signature by policy year, per policy issued at outset.

Hence calculate the expected present value at time 0 of the non-unit profits. [4]

iv) Calculate:

- EPV of future profits without non unit reserves
- EPV of future profits with non unit reserves

Comment briefly on why these two EPV of future profits may differ. [3]

v) The insurer is considering adding a rider for future new business under which the policy would cover two independent lives, both aged 35 exact, and on the first death during the 3-year term an additional non-unit payment of Rs. 50,000 would be made at the end of the year of death.

Assuming both lives follow the mortality basis given above, calculate the expected present value at time 0 of the additional rider outgo.

Comment briefly on the effect of this rider on:

- the non-unit reserve, and
- new business strain.

[5]
[20 marks]
